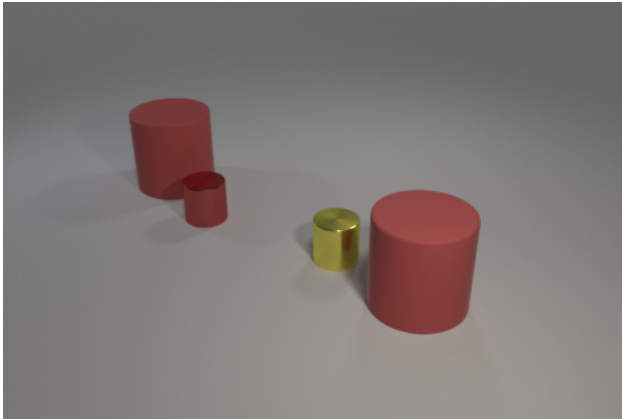




What is the color of the metal object to the left of the tiny yellow cylinder?

```

1 {'directions': {'above': [0.000, 0.000, 1.000],
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3                     'below': [-0.000, -0.000, -1.000],
4                     'front': [0.754, -0.656, -0.000],
5                     'left': [-0.656, -0.754, 0.000],
6                     'right': [0.656, 0.754, -0.000]},
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8   'image_index': 115,
9   'objects': [{ '3d_coords': [-2.447, -1.289, 0.700],
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13                  'rotation': 118.645,
14                  'shape': 'cylinder',
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17                  'color': 'yellow',
18                  'material': 'metal',
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20                  'rotation': 55.833,
21                  'shape': 'cylinder',
22                  'size': 'small'},
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24                  'color': 'red',
25                  'material': 'metal',
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27                  'rotation': 31.723,
28                  'shape': 'cylinder',
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33                  'pixel_coords': [321, 198, 9.103],

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34         'rotation': 45.535,
35         'shape': 'cylinder',
36         'size': 'large']],
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38                       'front': [[1, 2, 3], [3], [1, 3], []],
39                       'left': [[], [0, 2], [0], [0, 1, 2]],
40                       'right': [[1, 2, 3], [3], [1, 3], []]},
41     'split': 'val'}

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3  'image_index': 115,
4  'program': [{ 'function': 'scene', 'inputs': [], 'value_inputs': []},
5               { 'function': 'filter_size',
6                 'inputs': [0],
7                 'value_inputs': ['small']}],
8               { 'function': 'filter_color',
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10                'value_inputs': ['yellow']}],
11              { 'function': 'filter_shape',
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13                'value_inputs': ['cylinder']}],
14              { 'function': 'unique', 'inputs': [3], 'value_inputs': []},
15              { 'function': 'relate', 'inputs': [4], 'value_inputs': ['left',
16                               ]}],
16              { 'function': 'filter_material',
17                'inputs': [5],
18                'value_inputs': ['metal']}],
19              { 'function': 'unique', 'inputs': [6], 'value_inputs': []},
20              { 'function': 'query_color', 'inputs': [7], 'value_inputs': []
21                ]}],
22  'question': 'What is the color of the metal object to the left of the
23              tiny '
24              'yellow cylinder?',
25  'question_family_index': 75,
26  'question_index': 1156,
27  'split': 'val'}

```

Type associated with question:

$$\left[\begin{array}{ll}
 x & :Ind \\
 x_{\text{size}} & : \langle \lambda v : Ind . \text{small}(v), \langle x \rangle \rangle \\
 x_{\text{color}} & : \langle \lambda v : Ind . \text{yellow}(v), \langle x \rangle \rangle \\
 x_{\text{shape}} & : \langle \lambda v : Ind . \text{cylinder}(v), \langle x \rangle \rangle \\
 y & :Ind \\
 y_{\text{material}} & : \langle \lambda v : Ind . \text{metal}(v), \langle y \rangle \rangle \\
 c & : \langle \lambda v_1 : Ind . \lambda v_2 : Ind . \text{left}(v_1, v_2), \langle x, y \rangle \rangle
 \end{array} \right]$$

Witness record (extracted from scene graph):

$$\left[\begin{array}{l}
\begin{array}{l}
x=115_{-1} \\
\left. \begin{array}{l}
\text{color}=\text{yellow} \\
\text{size}=\text{small} \\
\text{rotation}=55.83304526084006 \\
\text{shape}=\text{cylinder} \\
3d_{\text{coords}}=[1.3542927503585815, -0.7202857136726379, 0.3499999940395355] \\
\text{material}=\text{metal} \\
\text{pixel}_{\text{coords}}=[256, 182, 10.231954574584961] \\
\text{id}=115_{-1} \\
\text{right}=[115_{-3}] \\
\text{behind}=[115_{-0}, 115_{-2}] \\
\text{front}=[115_{-3}] \\
\text{left}=[115_{-0}, 115_{-2}]
\end{array} \right\} \\
x_{\text{size}}=\langle \left. \begin{array}{l}
\text{color}=\text{yellow} \\
\text{size}=\text{small} \\
\text{rotation}=55.83304526084006 \\
\text{shape}=\text{cylinder} \\
3d_{\text{coords}}=[1.3542927503585815, -0.7202857136726379, 0.3499999940395355] \\
\text{material}=\text{metal} \\
\text{pixel}_{\text{coords}}=[256, 182, 10.231954574584961] \\
\text{id}=115_{-1} \\
\text{right}=[115_{-3}] \\
\text{behind}=[115_{-0}, 115_{-2}] \\
\text{front}=[115_{-3}] \\
\text{left}=[115_{-0}, 115_{-2}]
\end{array} \right\} \\
x_{\text{color}}=\langle \left. \begin{array}{l}
\text{color}=\text{yellow} \\
\text{size}=\text{small} \\
\text{rotation}=55.83304526084006 \\
\text{shape}=\text{cylinder} \\
3d_{\text{coords}}=[1.3542927503585815, -0.7202857136726379, 0.3499999940395355] \\
\text{material}=\text{metal} \\
\text{pixel}_{\text{coords}}=[256, 182, 10.231954574584961] \\
\text{id}=115_{-1} \\
\text{right}=[115_{-3}] \\
\text{behind}=[115_{-0}, 115_{-2}] \\
\text{front}=[115_{-3}] \\
\text{left}=[115_{-0}, 115_{-2}]
\end{array} \right\} \\
x_{\text{shape}}=\langle \left. \begin{array}{l}
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\text{size}=\text{small} \\
\text{rotation}=55.83304526084006 \\
\text{shape}=\text{cylinder} \\
3d_{\text{coords}}=[1.3542927503585815, -0.7202857136726379, 0.3499999940395355] \\
\text{material}=\text{metal} \\
\text{pixel}_{\text{coords}}=[256, 182, 10.231954574584961] \\
\text{id}=115_{-1} \\
\text{right}=[115_{-3}] \\
\text{behind}=[115_{-0}, 115_{-2}] \\
\text{front}=[115_{-3}] \\
\text{left}=[115_{-0}, 115_{-2}]
\end{array} \right\} \\
y=115_{-2} \\
\left. \begin{array}{l}
\text{color}=\text{red} \\
\text{size}=\text{small} \\
\text{rotation}=31.72266626640679 \\
\text{shape}=\text{cylinder} \\
3d_{\text{coords}}=[-0.9759458899497986, -1.5626593828201294, 0.3499999940395355] \\
\text{material}=\text{metal} \\
\text{pixel}_{\text{coords}}=[155, 151, 11.327897071838379] \\
\text{id}=115_{-2} \\
\text{right}=[115_{-1}, 115_{-3}] \\
\text{behind}=[115_{-0}] \\
\text{front}=[115_{-1}, 115_{-3}] \\
\text{left}=[115_{-0}]
\end{array} \right\} \\
y_{\text{material}}=\langle \left. \begin{array}{l}
\text{color}=\text{yellow} \\
\text{size}=\text{small} \\
\text{rotation}=55.83304526084006 \\
\text{shape}=\text{cylinder} \\
3d_{\text{coords}}=[1.3542927503585815, -0.7202857136726379, 0.3499999940395355] \\
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\text{pixel}_{\text{coords}}=[256, 182, 10.231954574584961] \\
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\text{right}=[115_{-3}] \\
\text{behind}=[115_{-0}, 115_{-2}] \\
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\text{left}=[115_{-0}, 115_{-2}]
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c=\langle \left. \begin{array}{l}
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\text{shape}=\text{cylinder} \\
3d_{\text{coords}}=[-0.9759458899497986, -1.5626593828201294, 0.3499999940395355] \\
\text{material}=\text{metal} \\
\text{pixel}_{\text{coords}}=[155, 151, 11.327897071838379] \\
\text{id}=115_{-2} \\
\text{right}=[115_{-1}, 115_{-3}] \\
\text{behind}=[115_{-0}] \\
\text{front}=[115_{-1}, 115_{-3}] \\
\text{left}=[115_{-0}]
\end{array} \right\}
\end{array} \right]
\end{array}$$